

Violence Reduction and Access to Higher Education

Post-Conflict Evidence from Colombia

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May 8th, 2026

Motivation

THEORETICAL FRAMEWORK AND GAP

Human capital theory

Education raises income and productivity, promoting development and social mobility. Returns to higher education are significant and grow over time (Becker 2002; Psacharopoulos & Patrinos 2018).

Conflict disrupts human capital accumulation

Armed conflict reduces enrolment, increases dropout, and lowers performance — documented in Iraq, Bosnia, Peru, Rwanda, and Colombia (Jürges et al. 2022; Soler 2016; Fergusson et al. 2020).

Violence reduction — unknown effect on higher education

Prior work shows that peace can aid educational recovery at primary/secondary levels (Prem et al. 2023; Montaña-Bardales & Palacios 2025), but no evidence exists for higher education.

RESEARCH QUESTION

How does the reduction of violence following the Colombian peace agreement affect access to higher education in municipalities most exposed to FARC violence?

TWO PLAUSIBLE CAUSAL MECHANISMS

1 Security ↑

Violence reduction allows youth to complete secondary school and physically access HE institutions (Peña & Dorussen 2021).

2 Academic performance ↑

Improved test scores and graduation rates post-ceasefire strengthen HE admission eligibility (Prem et al. 2023 and Montaña-Bardales & Palacios, 2025).

Context: Why Colombia?

CONFLICT TIMELINE

- **1964**
FARC & ELN founded — Ideological conflict inspired by communism and Cuba revolution
- **1980s**
AUC paramilitaries emerge; drug trafficking transforms the conflict
- **2012**
Peace negotiations with the FARC guerrilla began in Havana - Cuba
- **Dec 2014**
A bilateral ceasefire between FARC and the Colombian Government was declared
- **Nov 2016**
Signing of the final Peace Agreement

HUMAN COST 1995–2018

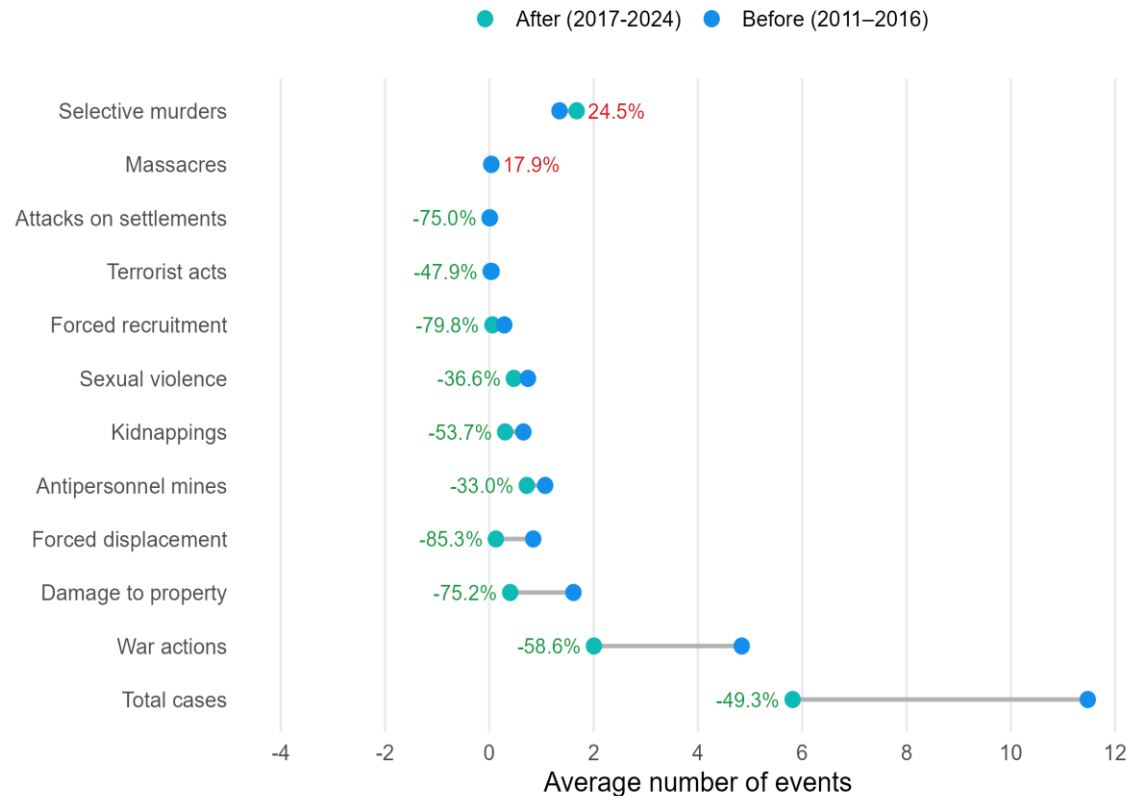
450,664	752,964	121,768	50,770
lives lost	displaced	disappeared	kidnapped

WHY THIS CASE IS IDEAL

- The longest conflict in the Western Hemisphere that has been extensively studied, but higher education is unexplored
- Peace agreement provides a clear exogenous shock for causal identification
- Rich municipal and individual-level data from public institutions
- Prior studies confirm effects on primary/secondary, and this study will extend the evidence upward

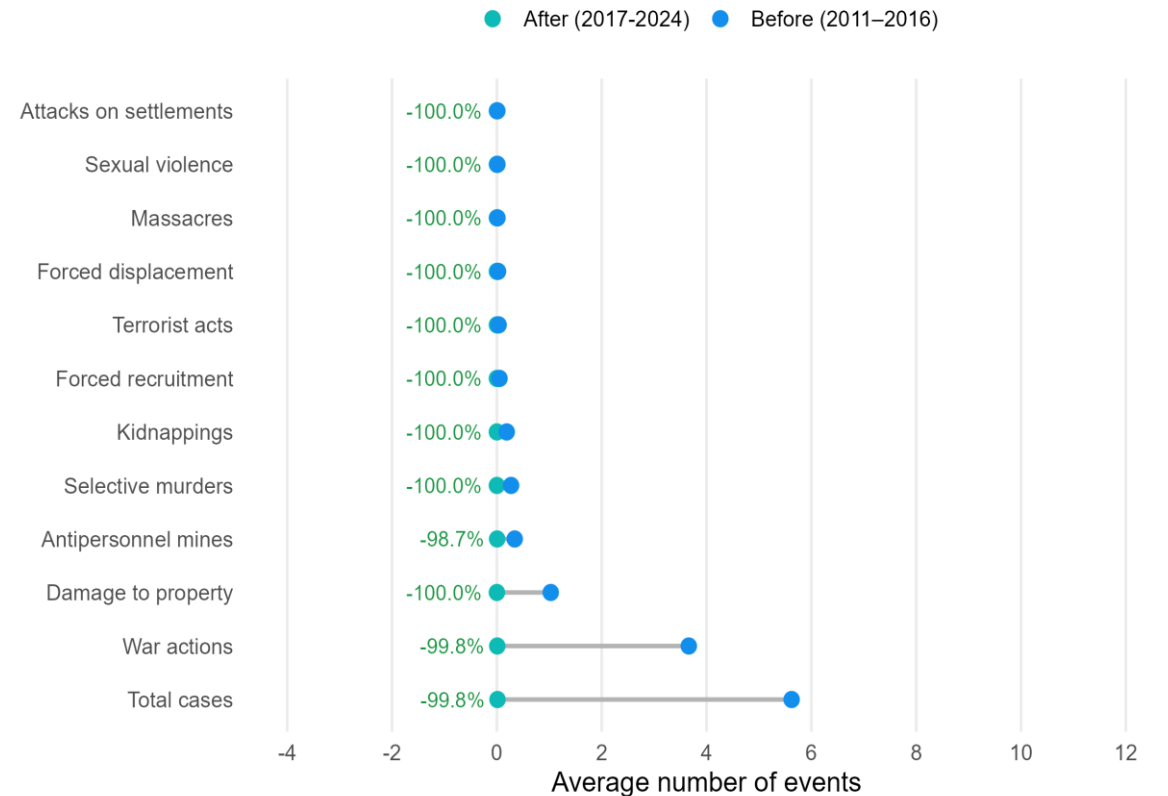
Context: Has violence declined?

Before vs After Violence Metrics
Comparison of average events by category



Source: Author's calculations based on conflict event data from the Colombian National Center of Historical Memory. The figure reports the average annual number of violent events related to civil conflict by category before (2011–2016) and after (2017–2024) the 2016 peace agreement.

Before vs After FARC-Related Violent Events
Comparison of average events by category



Source: Author's calculations based on conflict event data from the Colombian National Center of Historical Memory. The figure reports the average annual number of FARC-related events by category before (2011–2016) and after (2017–2024) the 2016 peace agreement.

Research Design

IDENTIFICATION STRATEGY

Difference-in-Differences (DiD)

- The DiD compares changes in the rate of transition to higher education between treated and control municipalities, before and after the 2016 peace agreement.
- The peace agreement provides a clear before-and-after threshold and well-defined treatment groups based on pre-existing FARC presence.
- DiD differences out municipality-level time-invariant factors (via municipality FE) and common temporal shocks (via year FE).

Key assumption

Parallel trends: absent the agreement, transition rates in treated and control municipalities would have followed similar trajectories.

No anticipation: treated municipalities did not adjust behaviour before the agreement took effect.

DATA

Outcome

ICFES – NATIONAL MINISTRY OF EDUCATION 2011-2024

Transition rate to higher education: proportion of secondary graduates in each municipality who enroll in a HE (technical, technological, or university) the following year.

Annual, municipal level, based on individual data.

Treatment

NATIONAL CENTER OF HISTORICAL MEMORY – CNMH 2011 - 2024

FARC-related violent events per municipality. Treatment dummy = 1 for municipalities where FARC events per capita are in the top three quartiles during 2011–2016, followed Prem et al. 2023.

Controls

ADMINISTRATIVE DATA – NATIONAL DEPARTMENT OFFICE

Tax income, national transfers, public expenditure, and hectares under coca cultivation.

Panel: 1,122 municipalities × 2011–2024

Challenges and limitations

FARC-only scope

The agreement covers FARC exclusively. ELN, paramilitaries, and successor groups continue operating, and their violence may attenuate or confound the estimated effect.

→ Include the other conflict actors' activity index as a control variable

Treatment measure

A single threshold to classify high/low FARC exposure loses within-group intensity variation. Different cutoffs (median, percentiles) may produce different results.

→ Robustness checks with multiple thresholds and a continuous exposure measure

Confounding educational policies

The HE access programmes (Ser Pilo Paga, Generacion E) overlap with the treatment period. Year FE absorbs common shocks, but differential program penetration in treated municipalities would confound the estimate.

→ Check available information to control for the program incidence per municipality.

Parallel trends and spillovers

The key DiD assumption must be validated empirically. Additionally, violence reduction in treated municipalities may affect neighbours (SUTVA concern), contaminating the control group.

→ I will check treated municipalities and their locations.

Next Steps

1

Violence measure and descriptive analysis

- Construct FARC exposure measure at multiple thresholds
- Map treated vs. control municipalities geographically
- Plot pre-treatment trends visually (2011–2016)
- Summary statistics by treatment group

2

Main Estimation & Robustness

- Two-way FE DiD with municipality and year FE, SE clustered at the municipality level
- Alternative violence thresholds (median, percentiles) and continuous exposure
- Sensitivity to treatment date (2014 ceasefire vs. 2016)

3

Extensions

- Heterogeneity analysis by gender
- ELN and coca hectares as additional controls

Capstone plan

Time: Mid may – Last week of June

Data construction

- Armed conflict database **OK**
 - *Treatment measure*
- Higher education access measure per municipality. **Work in progress**
- Control variables per municipality.
- Construct municipal panel 2011-2024

It will be ready by:

2 weeks – First week of June
23rd may – 6th June

Descriptives & pre-trends

- Treated vs. control map of municipalities
- Summary statistics by treatment group
- Raw time-series of transition rates by treatment group, 2011–2024
- Event-study plot — leads and lags around 2016 to test parallel trends assumption.

1 week– Second week of June
7th -13th June

Main DiD estimation

- Two-way FE: municipality and year, SE clustered at municipality
- Three specs: outcome only / + controls / + specification with interaction between other armed conflict actors and treatment variable

2 weeks – Third and fourth weeks of June : 14th -27th